

The Synthetic Nomological Net: A search engine to identify conceptual overlap in measures in the behavioral sciences

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Online supplement:

- Preprint: <https://osf.io/preprints/psyarxiv/nfgmv>
- GitHub: <https://github.com/synth-science/synth-net/>
- App: <https://huggingface.co/spaces/magnolia-psychometrics/synth-net>

Abstract:

Every month, scores of new measures and constructs enter the behavioral science literature. Researchers struggle to keep an overview of their subfields, let alone the wider field, with an estimated number of allegedly different constructs now exceeding 30,000. Proposing constructs is easy, finding redundancies and overlaps is not. The effort of finding redundancies using participant responses grows quadratically with the number of measures. This cost motivates the use of language models, which can simplify and speed up the process without requiring new data collection. Earlier work employed latent semantic analysis (LSA) to make use of the textual nature of questionnaires in the social and behavioral sciences. LSA's reliance on word co-occurrences, however, can produce misleading similarity estimates that fall short of accuracy and may even reproduce the very jingle-jangle fallacies it aims to detect. Recent advances demonstrate that transformer-based sentence embedding models offer a more viable solution for predicting empirical relatedness

between survey items. In the presented work, we use a fine-tuned language model, the SurveyBot3000, to introduce the Synthetic Nomological Net, an open-access web application for systematic detection of conceptual overlap. The application indexes over 470,000 items across 74,000 scales from approximately 31,500 instruments in the APA PsycTests database.

Introduction

The steady growth of proposed constructs and measures in the behavioral science literature has come at the cost of taxonomic coherence. When Cronbach and Meehl (1955) envisioned the nomological network to map out the relationships between psychological constructs and their corresponding manifestations in the physical world, they noted that scientists have the freedom to expand the network, if an observation will not fit into it. Nowadays, behavioral scientists indeed frequently expand the network, but they often neglect to first show that an observation did not fit. The reality is that over 30,000 allegedly novel constructs have been proposed with new additions being introduced each month (Anvari et al., 2025; Elson et al., 2023). Proliferation departs from the scientific conservatism Cronbach and Meehl advocated and has led to a fragmentation of the field, limiting the generation of cumulative knowledge. Among the most problematic consequences of this fragmentation are jingle-jangle fallacies: cases where researchers unwittingly study the same construct under different names (jangle fallacies; e.g., grit and conscientiousness, Credé et al., 2017) or conflate distinct constructs under a single label (jingle fallacies; e.g., mindfulness, Altgassen et al., 2024).

Conventional approaches to evaluating construct redundancy cannot keep pace with this proliferation. For example, the Synthetic Aperture Personality Assessment (SAPA; Condon & Revelle, 2015) administers random subsets of items from various personality scales to survey respondents. With growing sample size, SAPA-data can effectively be used to evaluate the (non-)redundancy of measures and proposed constructs. Unfortunately, this approach doesn't scale well, as including additional measures requires a quadratic increase in survey participants. To obtain a pairwise N of 100 for correlations between 470,000 items in a SAPA study in which each participant answers 100 items, we would need more than 2 trillion participants.

Given the textual nature of surveys, questionnaires, checklists, tests and other forms of assessment in the social and behavioral sciences, natural language processing holds the potential to address the inconsistent mapping between measures and constructs. Early works in the field have demonstrated that survey item semantics are related to survey responses (e.g., Arnulf et al., 2014; Larsen & Bong, 2016) using latent semantic analysis (LSA; Deerwester et al., 1990). In essence, LSA maps text data (i.e., survey items) into a semantic space by identifying latent topics that underlie word co-occurrences across documents (i.e., scales). Numerical vector representations of items and scales can then be obtained and compared using distance metrics, such as cosine similarity. Building on LSA, Rosenbusch et al. (2020) developed the Semantic Scale Network, an online tool to

automatically detect semantic overlap between psychological scales. The application allows researchers to input items from a new scale and receive quantifications of semantic similarity with over 4,000 existing scales in its corpus, without requiring empirical data collection. The Semantic Scale Network first showed the potential of using natural language processing to detect potentially redundant measures in the scientific literature. However, a central limitation arises from the reliance on word co-occurrences of LSA-based approaches. As noted by Rosenbusch and colleagues, semantic similarity between two conceptually overlapping scales may be low, if the items of each scale are phrased differently (e.g., “to have a lot of friends” and “to have plentiful companions”). Similarity estimates between scales may likewise appear inflated, if two items measuring different constructs employ similar wording (e.g., homonyms, as in “I am the life of the party” and “I vote for the same party in every election”). In that sense, LSA-based approaches may even reproduce jingle-jangle fallacies due to surface level characteristics of the item text.

Moving beyond the constraints of LSA, transformer models have emerged as the current state-of-the-art in many natural language processing applications (Vaswani et al., 2017; see Hussain et al., 2023 for an accessible introduction). In contrast to LSA, transformer-based embedding models produce contextualized n-dimensional vector representations of text, with each token (i.e., word or sub-word) encoded according to its role within a particular sentence. For example, the aforementioned usage of the word “party” would yield different embeddings depending on the context of each item. Rather than embedding individual tokens, sentence transformer models can be used to represent entire text sequences as a unified vector (Reimers & Gurevych, 2019). Current research demonstrates the predictive power of these embeddings for understanding survey responses. At the item level, Abdurahman et al. (2023) extrapolated participant ratings to novel personality items using sentence embeddings. At the aggregate level, researchers have trained models to produce item embeddings to predict pairwise correlations between items and scales, as well as reliability coefficients with high accuracy (Hernandez & Nie, 2023; Hommel & Arslan, 2025; Schoenegger et al., 2025; Wulff & Mata, 2025).

In summary, sentence embeddings offer a solid basis for detecting overlap among measures, because they a) capture contextualized meaning in text beyond shared wording, b) remain robust to paraphrasing and syntactic variations, c) can be produced to reflect the empirical association between survey scales, and d) scale efficiently to thousands of instruments without requiring additional data collection.

In this work, we introduce the Synthetic Nomological Net: a web application powered by the SurveyBot3000, a language model developed by Hommel and Arslan (2025). The Synthetic Nomological Net enables systematic detection of conceptual overlap across nearly 30,000 psychological measures from the APA PsycTests database.

The Synthetic Nomological Net

We release the Synthetic Nomological Net (SynthNet) as an open-access web application on Hugging Face Spaces: <https://huggingface.co/spaces/magnolia-psychometrics/synth-net>¹.

The web application functions as an ontological search engine, currently allowing researchers and practitioners to explore 29,715 instruments extracted from the APA PsycTests database, with permission of the American Psychological Association. In contrast to conventional search engines which adopt keyword-based approaches, SynthNet relies on sentence embeddings of items, scale labels and instrument names to match user queries with database records. The application operates in two different modes: item-based search and scale-based search.

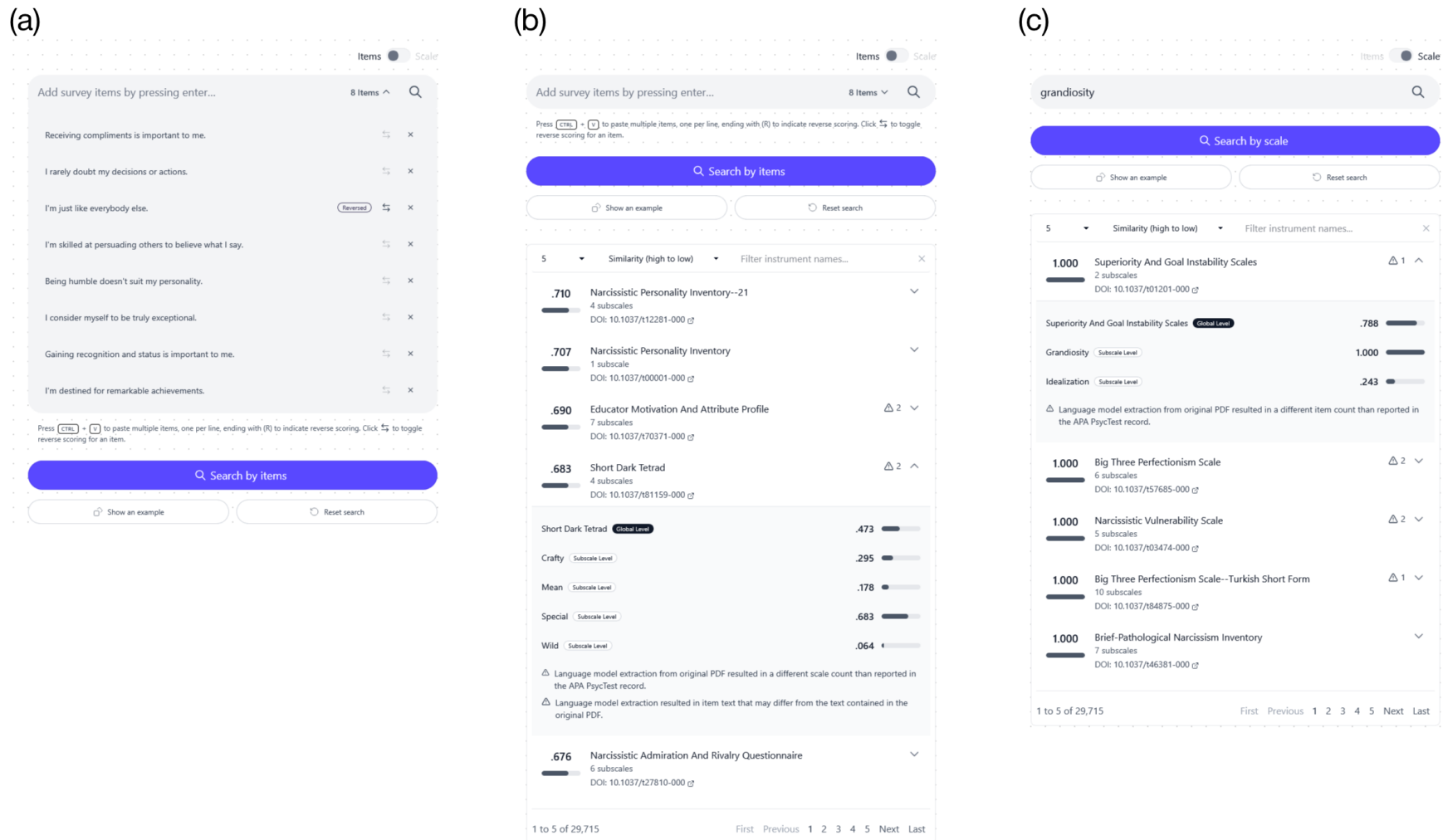
In item-based search, users enter one or more items into the search interface. Items can come from existing scales, scales under development, or be formulated ad hoc to explore a construct of interest. We recommend this mode for detecting conceptual overlap, as it operates on the SurveyBot3000 model, which was explicitly trained to predict pairwise correlations between items and scales (Hommel & Arslan, 2025).

The utility of item-based search is exemplified by the results depicted in Figure 1. The example shows item queries plausibly related to narcissism, with reverse-coded items appropriately flagged (Figure 1a). The resulting matches (Figure 1b) reveal substantial conceptual overlap with multiple instruments measuring dark personality traits. Notably, SynthNet identified high similarity with the Short Dark Triad (Paulhus et al., 2020), particularly its subscale labeled "special". Here, Paulhus and colleagues used a euphemistic label in place of the term narcissism. This illustrates how focusing on item content rather than scale labels allows SynthNet to identify substantive overlap that is missed by conventional search methods.

¹ Updated links will be maintained at <https://github.com/synth-science/synth-net> in case of migration. The web application's source code is freely available at <https://huggingface.co/spaces/magnolia-psychometrics/synth-net/tree/main>.

In scale-based search, users enter a construct, scale or instrument name (see Figure 1c). This mode uses a smaller, general purpose model and produces results which reflect semantic overlap, which is less likely to correspond to empirical relatedness in measures.

Figure 1. The Synthetic Nomological Net web application interface.



Note. The example shows how to use SynthNet to search for instruments related to narcissism in a) item-based search, and c) scale-based search.

Below, we detail the development of SynthNet, including its underlying models, algorithms and database.

Sentence Embedding Models

To transform the extracted items and scale labels into searchable representations, we employed two distinct sentence embedding models: The SurveyBot3000 for item-based search and the all-MiniLM-L6-v2 for scale-based search.

The SurveyBot3000 builds on the all-mpnet-base-v2 sentence transformer model, which Hommel & Arslan (2025) fine-tuned through a two-stage training protocol to predict pairwise item correlations. In the first stage, the model was calibrated to distinguish semantically opposing concepts, ensuring items measuring opposite construct poles (e.g., extraversion versus introversion) produce negatively correlated embeddings. This calibration was necessary because standard sentence transformers typically yield positive cosine similarities even for contradictory statements, as their high-dimensional representations encode multiple correlated linguistic features. In the second stage, the model was fine-tuned on 74,339 item pairs from various questionnaires using a cosine similarity loss objective. Here, the model learned to predict item relatedness from the text in the item stems by minimizing the discrepancy between predicted and empirical correlations. Initial validation on a hold-out sample demonstrated strong convergence between synthetic and empirical correlations for item pairs, scale pairs, and Cronbach's alpha coefficients. A subsequent pre-registered validation study confirmed generalization to new content domains, achieving out-of-sample accuracies of $r = .59$ for item correlations, $.84$ for scale correlations, and $.84$ for reliability coefficients. Because of these strong predictive capabilities, we use the SurveyBot3000 to infer correlations between scales in the APA PsycTests data and thereby to identify potential conceptual overlap.

In addition to the SurveyBot3000 model, SynthNet uses the smaller all-MiniLM-L6-v2² model to power scale-based search. Here, in contrast to item-based search, scale-representations in SynthNet are constructed by computing 384-dimensional embeddings from construct labels and instrument names.

Employing both models allows researchers to probe the SynthNet for different kinds of potential overlap between search queries and existing measures in APA PsycTests, i.e., on the conceptual and on the semantic level.

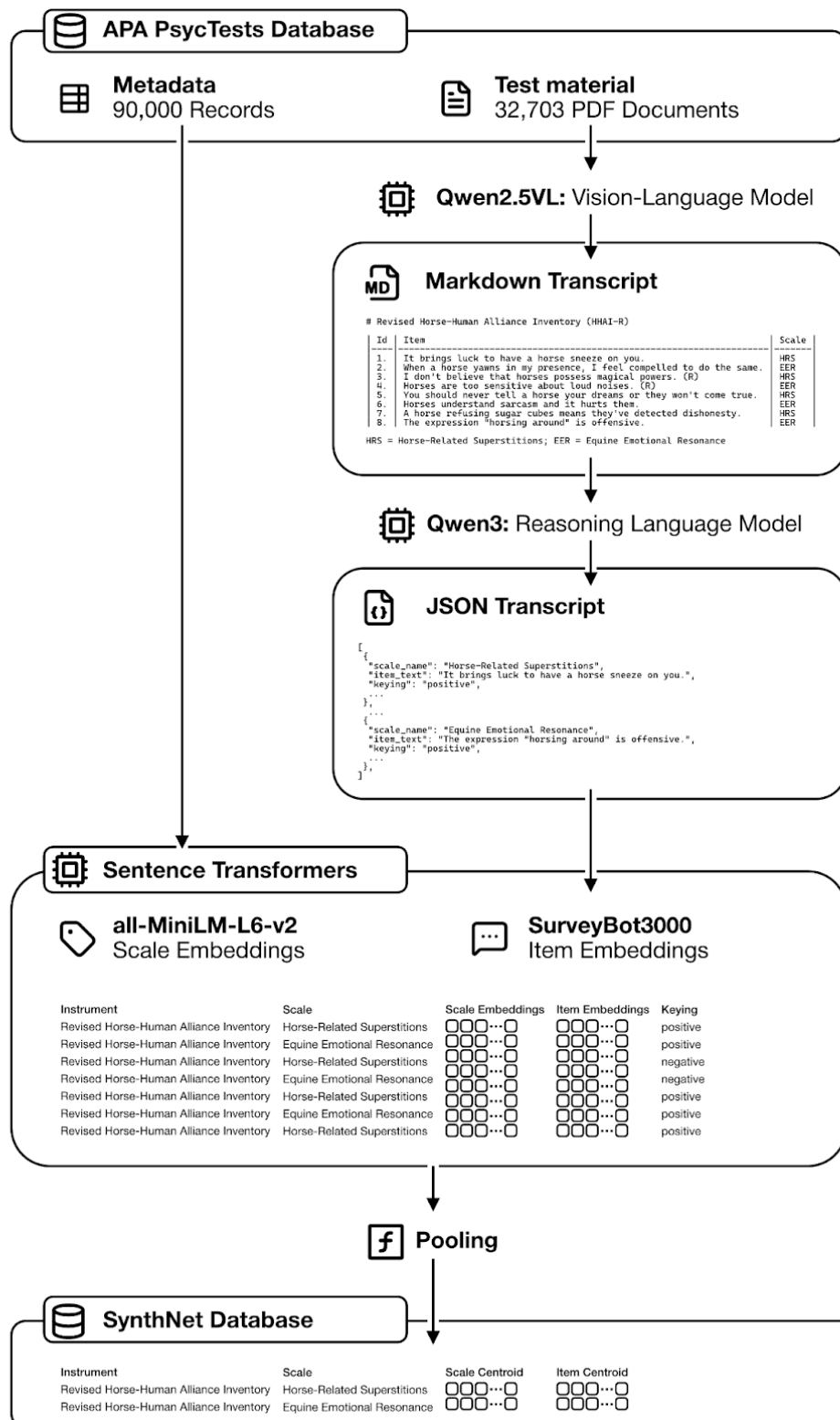
² <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2> (retrieved April 1, 2025)

Information Extraction from PsycTest Documents

The PsycTests database, maintained by the American Psychological Association, constitutes one of the largest collections of measures in the social and behavioral sciences, currently containing over 90,000 records with monthly additions. Each record includes metadata such as constructs measured, instrument type, associated references, and publication year. Where available, records also include the actual test materials and supporting documentation.

SynthNet is built on a PsycTests database snapshot obtained on August 21, 2023, which contained 32,703 PDF documents of test materials. Documents in PsycTests consist of unstructured content, combining textual and image data from scanned pages. As depicted in Figure 2, preprocessing the documents involved a two-step information extraction method to parse the data into a structured format by using two open-source transformer-based language models within the ollama framework for local computing (*Ollama*, 2025).

Figure 2. Overview of the SynthNet Data Extraction and Embedding Pipeline



Note. Two-stage extraction and embedding pipeline used to process APA PsycTests documents into structured data for SynthNet, including vision-language transcription, reasoning-based structuring, embedding generation, and database integration.

First, we used a vision-language model (Qwen2.5-VL; Bai et al., 2025) to convert document pages into Markdown text format. While traditional optical character recognition (OCR) techniques perform this conversion mechanistically, character by character, transformer-based vision-language models interpret visual input contextually. Contextual processing enables the model to compensate for obscured characters or poor image quality by making use of surrounding semantic information, and thereby improving extraction accuracy for degraded or low-quality documents. A limitation of this approach is the potential for confabulations, wherein the model may generate plausible but factually incorrect content when interpreting ambiguous or unclear visual information (Liu et al., 2024).

In the second step of the extraction process, we parsed and structured the Markdown text into a standardized format using a 32-billion parameter generative reasoning language model (Qwen3; Yang et al., 2025) with prompt-based instructions. Compared to non-thinking models, reasoning models achieve better performance at complex tasks by generating intermediate reasoning steps before producing a final response. We instructed Qwen3 to extract each item from the instrument along with its associated scale label and keying information, and to output the structured data in JSON format. As APA PsycTests documents varied in abundance of information, some content was inferred by the model. For example, it frequently was the case that the documents merely stated the instrument names and items, but were not explicit on the measured construct. Specific details of the extraction process are provided at: https://github.com/synth-science/synth-net/tree/main/document_extractor

The entire extraction process required 61 days of continuous computation on two NVIDIA RTX A6000 GPUs with 96 GB of VRAM. Because the sentence embedding models employed in SynthNet have limited multilingual capabilities, we reduced the data to English-language instruments. Given inconsistencies in PsycTests language metadata, we excluded records with non-English instruments, though complete certainty regarding the language of all retained records cannot be guaranteed. From the remaining 29,715 instruments, the extraction processes resulted in 74,372 scales with 473,263 items in total.

Calculating Embeddings

We constructed a hierarchical embedding system that represents instruments at both global and subscale levels. For instruments containing multiple subscales, we generated separate embeddings for each subscale in addition to an overall instrument-level

embedding. In doing so, we did not take into account whether the instrument measured a coherent construct (e.g., extraversion) or comprised theoretically independent constructs (e.g., a Big Five instrument). The specific embedding calculation procedure varied depending on the search mode.

In item-based search, we began by encoding each individual item with the SurveyBot3000, producing 768-dimensional vector representations. The item-level embeddings were then aggregated to produce scale-level representations through mean pooling.

Conventional scale analysis requires reverse-coding negatively-keyed items before computing scale scores. Likewise, SurveyBot3000 embeddings of opposite-pole items correlate negatively and must be reversed before aggregation. To this end, we first averaged embeddings for positively- and negatively-keyed items separately to produce polar centroids. The polarity axis, computed by subtracting the negative centroid from the positive centroid, represents the primary dimension of construct variation. We then positioned a hyperplane at the midpoint between the centroids, which separates the embedding space into positive and negative regions. Negatively-keyed items can be reversed by reflecting them across this plane. The reflection is computed by projecting each item onto the polarity axis, measuring its signed distance from the hyperplane, and shifting it an equal distance to the opposite side. We applied this reflection to items that were both a) explicitly marked as reverse-coded in the original instrument and b) synthetically oriented toward the negative polar centroid (i.e., showing negative cosine similarity with the positive centroid). The python implementation of this algorithm can be found in the online supplemental section.

Scale-based search employed a simpler approach. We used the all-MiniLM-L6-v2 model to separately encode the instrument name (obtained from PsycTests metadata) and the scale name (inferred by Qwen3), then averaged these two embeddings to produce a single 384-dimensional vector representation.

Scoring Algorithm

Identifying conceptually related documents through embedding-based retrieval, commonly termed semantic search (Reimers & Gurevych, 2019), constitutes a standard application of sentence transformers. The retrieval mechanism embeds documents (i.e., instruments and scales) and search queries into a shared semantic space. For the SurveyBot3000 model, this space approximates nomological relationships between

constructs. Relatedness is then quantified using cosine similarity, which captures the angular proximity between embeddings.

In item-based search, users specify their query by entering survey items that purportedly measure a construct of interest, along with keying information specifying each item's polarity. Following the embedding procedure described previously, these query items are encoded using SurveyBot3000 and aggregated through polarity-aligned mean pooling. SynthNet then computes the cosine similarity between the resulting query embedding and all instrument- and subscale-level embeddings in the database. Scales exhibiting the highest absolute cosine similarity are therefore likely to measure constructs conceptually related to the queried scale. Scale-based search follows the same retrieval logic. Here, users provide a single keyword describing a construct, scale, or instrument name, which is embedded via all-MiniLM-L6-v2 and compared to the corresponding scale embeddings in the database.

Data Quality

In addition to the aforementioned minor inconsistencies in the APA PsycTests metadata, language-model extraction in preprocessing introduced data that was occasionally incomplete and sometimes erroneous. Such errors can arise when language models confabulate plausible sounding but factually incorrect responses, commonly termed "hallucinations" (e.g., Huang et al., 2025; Liu et al., 2024). Additionally, Qwen3 occasionally erred when inferring information left implicit in the documents. For instance, keying information specifying which items should be reverse-scored was documented inconsistently across instruments, potentially leading the model to overlook or misinterpret such specifications.

Quantifying these types of errors reliably is not feasible due to the scope and unstructured nature of the APA PsycTests database content. Nevertheless, to address uncertainty arising from potential extraction errors, SynthNet provides three metrics that flag potential inconsistencies between extracted data and source materials. These heuristic checks serve as signals for users to exercise additional scrutiny, rather than as definitive indicators of poor data quality. SynthNet reports a) divergence between the number of items extracted by Qwen3 and item counts documented in PsycTests metadata, b) discrepancies in scale counts between extraction and metadata specifications, and c) substantial deviations in extracted item text from the raw PDF content, identified through fuzzy string matching.

We deliberately implemented conservative heuristics that prioritize sensitivity over specificity. By accepting a higher rate of false positives, we minimize the risk of overlooking genuine extraction errors and ensure that SynthNet users are alerted even when the extraction may in fact be correct but metadata incomplete or ambiguous. Item count discrepancies were most common and affected 30.4% of instruments with available metadata. Deviations in item count can arise from various reasons such as incomplete or inaccurate metadata documentation, model hallucinations, or failure to parse complex document layouts. Scale count deviations were less prevalent and merely occurred in 5.17% of cases, but scale counts were also rarely recorded in the source data. Fuzzy string matching showed a high concurrence between extracted items and the text in the original documents, with an average overlap of 86.35% ($SD = 21.61\%$), with a heavily skewed distribution towards high similarity. However, 11,992 instruments (40.35%) contained at least one extracted item that did not meet the threshold of 95% string similarity. Such cases can arise when source documents contained scanned images or malformed layouts (e.g., table elements disrupting sequential item order). Low similarity can also result from model rephrasing or hallucinations. Consequently, 15,880 instruments (53.4%) were flagged for additional user scrutiny. We think the search results will be enhanced by these flags, providing the user with additional information on data quality to facilitate informed decisions.

Discussion

We introduce the Synthetic Nomological Net (SynthNet) to provide researchers and practitioners with an automated approach to locating measures within an increasingly sprawling ecosystem that has outgrown manual cataloging efforts. In contrast to conventional methods that rely on empirical data, SynthNet evaluates the relatedness of measures based on the textual information contained in items, scale and instrument names. The use of natural language processing to identify overlapping measures has precedent in tools like the Semantic Scale Network (Rosenbusch et al., 2020). SynthNet contributes several key improvements to this line of work. First, it employs sentence transformer models rather than LSA, producing contextualized embeddings that minimize spurious matches driven by coincidental wording and superficial linguistic aspects. Second, its use of the SurveyBot3000 model extends the capabilities of baseline sentence transformers in identifying conceptual relatedness, as the model is explicitly trained to approximate empirical relationships between measures (Hommel & Arslan, 2025). Third, by incorporating a large

portion of the APA PsycTests database, SynthNet provides access to one of the most comprehensive and authoritative collections of psychological measures.

Recommendations

SynthNet supports taxonomic refinement in two ways. Retrospectively, it enables systematic identification of redundancies in the existing literature. Prospectively, it allows researchers to screen for novelty before developing new measures. Importantly, SynthNet cannot and should not replace empirical investigations of nomological validity. We agree with Rosenbusch et al. that tools like SynthNet serve best as decision support systems, providing evidence to inform rather than replace expert judgment. Artificial intelligence, particularly language modeling techniques, is increasingly transforming meta-scientific practice. Concerns have been raised about potential threats to scientific integrity (Binz et al., 2025) and the risk of amplifying existing malpractices through unreflective use (Cummins, 2025). However, responsible human-in-the-loop applications can enhance research quality by addressing systemic problems. For instance, Cummins et al. (n.d.) proposed using language models to automatically flag deviations between preregistrations and submitted manuscripts, drawing reviewers' attention to potential discrepancies without replacing human judgment. We envision a similar role for SynthNet in the peer review process. When researchers propose new measures, SynthNet could help authors, editors, and reviewers quickly identify potential redundancies with existing instruments. Cases where SynthNet indicates high overlap with preexisting scales would require submitting researchers to justify the novelty of their measure through theoretical arguments and, ideally, empirical evidence of discriminant validity.

Limitations

SynthNet's results depend fundamentally on data quality. Despite considerable effort to ensure reliable automated extraction from APA PsycTests materials, errors remain inevitable. These include extraction failures, missing items, and occasionally hallucinated content. While we implemented quality metrics to alert users to potential issues, they cannot substitute for careful scrutiny of individual search results. Because source materials were poorly structured and often omitted information about nested scale structures, our extraction method only captured and subsequently calculated embeddings for scales on the instrument and subscale levels. This poses problems for search queries targeting hierarchically organized instruments, such as some Big Five inventories. For example, if users enter item queries that cover a broader measure of conscientiousness, they will be matched against a

global scale (i.e., “personality”) and subscales (e.g., “orderliness”) for instruments that reported subfacets. As a consequence, estimated overlap may appear deflated when the conceptual breadth of a query misaligns with the granularity of retrieved scales.

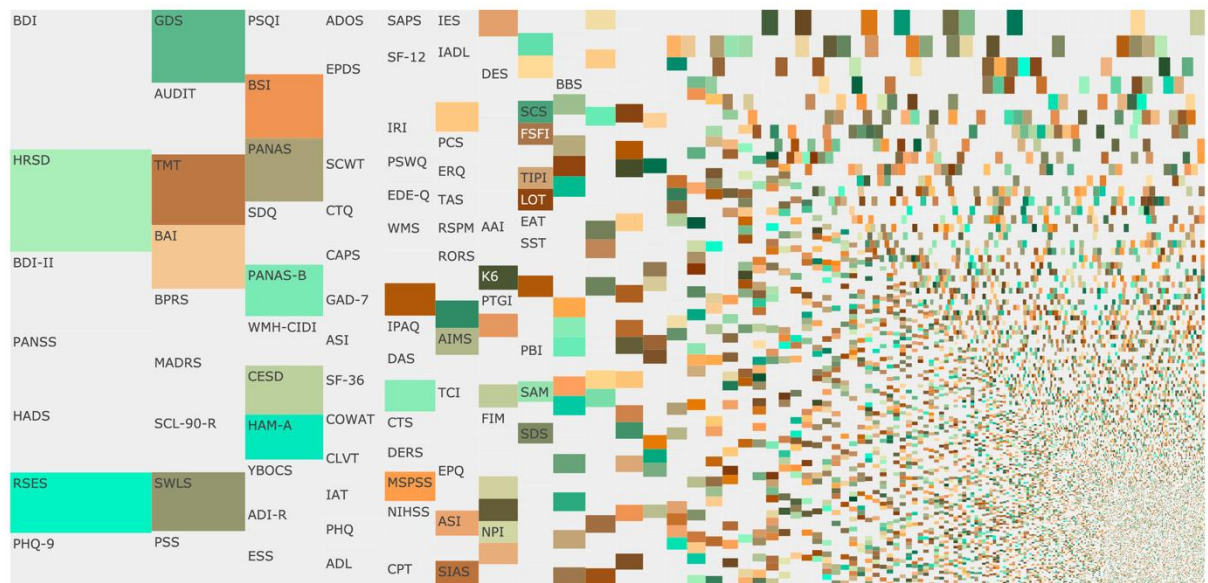


Figure 3. A treemap plot showing psychological measures in APA PsycTests. Each tile represents a measure. Tiles are sized proportionally to how often measures were documented to be used by APA coders in the PsycInfo database between 1993 and 2023. Only the measures shown in color had usable item text information in APA PsycTests and can be found using the SynthNet search. This treemap does not show tests that had no usage recorded in PsycInfo ($n = 40,548$).

Some additional minor constraints warrant mention. First, given the SurveyBot3000's limited multilingual capabilities, non-English or translated instruments that remain in the database may yield less accurate search results. Second, instruments containing atypical items (e.g., questionnaires with mixed response formats combining rating scales and open-ended questions) may yield less accurate results, as such formats deviate from the SurveyBot3000's training data. Third, because of the SynthNet lacks continuous integration with PsycTests, newly published instruments are not included in its database. Finally, legal constraints limit the information we can provide to users. We cannot display individual item text and must present results only at the aggregate scale or instrument level. These same constraints prevent us from linking to original publications, restricting us to providing DOI references to the corresponding APA PsycTests records.

Outlook and Future Research

The SynthNet data provides a strong foundation that can be expanded both temporally and in content breadth. Our August 2023 data snapshot can be updated to incorporate the steady stream of new measures published since then, ensuring researchers can screen for novelty against the most current literature. Beyond temporal expansion, coverage can extend to sources not fully represented in APA PsycTests, such as instruments found in supplementary materials, preregistration platforms, open science repositories, and adjacent behavioral science disciplines. Integration with large-scale data aggregation initiatives, such as the Item Response Warehouse (Domingue et al., 2025) and data donation platforms (e.g., (Reichmann & Hommel, n. d.)), would enable continuous database expansion and enhanced coverage. Parallel efforts should focus on improving structured data extraction from source materials, both by employing more capable vision-language and multilingual extraction models (e.g., Wei et al., 2025), and by implementing methods to mitigate extraction errors (e.g., semantic entropy; Farquhar et al., 2024).

Conclusion

A fundamental tension in the behavioral sciences is balancing researchers' freedom to propose new constructs and measures against the need to prevent unchecked proliferation that threatens taxonomic coherence. Previous tools attempting to strike this balance were constrained by methodological limitations and insufficient data coverage. SynthNet addresses both constraints, offering researchers a practical means to navigate the existing measurement landscape before adding to it. When researchers can efficiently map their work onto what already exists, construct proliferation transforms from a liability into an

opportunity for systematic refinement. As natural language processing capabilities advance and databases expand, we anticipate a shift from isolated measure development toward collaborative curation of an increasingly integrated nomological network.

Acknowledgments

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Data availability

The study relies on test materials obtained from the APA PsycTests database. In accordance with the licensing agreement with the American Psychological Association, we are not permitted to share the raw PDF documents or make item-level content publicly available. We provide the processing code and all reproducible components that can be shared under the terms of our agreement. Researchers with appropriate access to APA PsycTests can reproduce the dataset by applying our extraction and preprocessing scripts to their own licensed copy.

Code availability

We have shared the code for data extraction and preprocessing at <https://github.com/synth-science/synth-net>. The source code for SynthNet is shared at <https://huggingface.co/spaces/magnolia-psychometrics/synth-net/tree/main>.

Author contributions

B.E.H.: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Data Curation, Writing - Original Draft, Visualization, Supervision, Project administration, Funding acquisition.

A.I.K.: Validation, Formal analysis, Data Curation, Writing - Review & Editing, Visualization

R.C.A.: Conceptualization, Formal analysis, Resources, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Competing interests

Björn E. Hommel is affiliated with magnolia psychometrics GmbH, a private consulting agency which has agreed to maintain the SynthNet web application, currently hosted on Hugging Face. There are no competing interests, financial or otherwise, related to this research. Potential future commercial applications of these findings may be developed.

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